

WAINFLEET, United Kingdom

WMO: 033920

Lat: 53.0883N

Lon: 0.2708E

Elev: 16

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.0	(c) 30.1	(d) 23.2	(e) 17.5	(f) 31.0	(g) 25.7	(h) 19.7	(i) 32.5	(j) 35.7	(k) 45.9	(l) 32.6	(m) 46.5	(n) 8.8	(o) 270	(p) 0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.5	(c) 75.6	(d) 65.6	(e) 72.8	(f) 64.0	(g) 70.5	(h) 62.9	(i) 67.3	(j) 73.3	(k) 65.5	(l) 70.9	(m) 64.1	(n) 68.9	(o) 12.5	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	92.1	70.0	63.3	87.2	68.2	61.9	83.0	66.7	31.7	73.3	30.4	70.9	29.3	69.1	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 30.5	(b) 27.2	(c) 24.3	(e) 21.7	(f) 82.4	(g) 5.8	(h) 2.6	(i) 17.6	(j) 84.3	(k) 14.2	(l) 85.8	(m) 11.0	(n) 87.3	(o) 6.8	(p) 89.2		
			DB													
			WB													
				21.1	72.2	5.8	3.4	17.0	74.6	13.6	76.6	10.3	78.6	6.1	81.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	39.9	40.8	43.4	47.1	52.4	57.4	61.0	61.7	57.5	52.0	44.8	40.7
	DBStd	9.17	4.87	5.16	4.84	4.78	4.74	4.62	4.22	4.34	4.37	5.00	5.14	5.62
	HDD50	1423	315	259	209	111	29	2	0	0	2	36	168	292
	HDD65	5534	779	677	668	536	391	233	139	121	228	402	606	754
	CDD50	1402	0	2	6	25	103	224	340	362	227	99	12	2
	CDD65	39	0	0	0	0	0	4	14	18	3	0	0	0
	CDH74	166	0	0	0	0	3	29	58	65	9	2	0	0
	CDH80	12	0	0	0	0	0	3	4	5	0	0	0	0
	WSAvg	12.5	14.1	14.2	13.2	12.6	12.3	11.0	10.3	10.7	11.6	13.2	13.3	13.6
	PrecAvg	23.0	2.1	1.5	1.8	1.6	1.7	1.9	1.8	2.3	2.0	2.2	2.4	2.3
(15) Precipitation	PrecMax	30.6	4.3	4.6	4.5	3.9	3.7	4.8	5.4	5.5	4.7	5.2	4.5	5.7
	PrecMin	15.0	0.4	0.2	0.4	0.4	0.2	0.2	0.3	0.1	0.2	0.2	1.0	0.7
	PrecStd	4.0	0.9	0.9	0.9	0.8	0.8	1.2	1.1	1.2	1.2	1.4	1.0	1.1
	PrecStd	4.0	0.9	0.9	0.9	0.8	0.8	1.2	1.1	1.2	1.2	1.4	1.0	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	54.4	56.3	59.7	66.2	71.0	78.2	79.4	79.6	75.3	67.7	59.8	55.2	
	0.4% MCWB	52.1	52.8	53.3	57.9	62.7	67.5	67.4	67.8	65.8	61.0	57.2	52.8	
	2% DB	51.3	53.0	56.4	61.5	66.9	72.1	75.1	75.4	71.1	63.5	56.6	52.7	
	2% MCWB	49.2	50.0	51.3	54.8	59.7	63.8	64.8	65.6	63.5	59.2	54.4	50.8	
	5% DB	49.3	50.7	53.8	58.2	63.7	68.7	72.1	72.6	67.6	61.2	54.4	50.9	
	5% MCWB	47.4	47.9	49.6	52.5	57.2	61.5	63.8	63.8	61.2	57.8	52.6	49.2	
	10% DB	47.1	48.5	51.5	55.5	61.0	65.9	69.7	70.1	65.1	59.2	52.5	48.9	
	10% MCWB	45.4	46.2	47.9	50.8	55.3	59.7	62.6	62.7	59.8	56.3	50.8	47.3	
	0.4% WB	52.4	53.2	54.7	60.0	64.6	69.0	69.3	70.1	67.2	62.5	57.6	53.2	
	0.4% MCDB	54.1	55.4	58.1	64.2	69.3	76.2	76.3	76.6	72.7	66.7	59.0	54.5	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	49.7	50.6	52.3	56.1	60.5	65.2	66.8	67.1	64.4	60.2	54.9	51.2	
	2% MCDB	51.1	52.8	55.5	60.1	65.5	70.4	72.9	73.0	69.7	62.6	56.3	52.5	
	5% WB	47.6	48.4	50.2	53.5	58.3	62.6	65.1	65.5	62.5	58.6	52.8	49.5	
	5% MCDB	48.9	50.2	53.2	57.1	62.5	67.1	70.5	70.7	66.4	60.6	54.2	50.8	
	10% WB	45.7	46.6	48.4	51.4	56.4	60.8	63.6	64.0	60.9	56.9	50.9	47.6	
	10% MCDB	46.9	48.3	51.0	54.9	59.9	64.9	68.5	68.5	64.1	58.8	52.2	48.9	
(35) Mean Daily Temperature Range	MDBR	8.2	9.7	12.0	13.5	13.8	14.0	14.8	15.5	14.2	11.4	9.8	8.1	
	5% DB MCDBR	10.8	11.9	15.6	17.6	17.9	18.0	18.5	18.7	17.1	12.6	10.5	9.3	
	5% DB MCWBR	10.1	9.4	11.1	11.7	11.5	11.0	10.3	10.0	10.6	9.2	9.2	8.6	
	5% WB MCDBR	10.4	11.4	14.6	16.5	16.3	16.6	17.1	16.5	15.7	11.7	10.0	9.3	
	5% WB MCWBR	9.9	9.5	11.2	11.9	11.6	11.2	10.6	9.8	10.8	9.3	9.1	8.6	
	5% WB MCWBR	9.9	9.5	11.2	11.9	11.6	11.2	10.6	9.8	10.8	9.3	9.1	8.6	
(40) Clear-Sky Solar Irradiance	taub	0.328	0.343	0.376	0.402	0.402	0.409	0.410	0.398	0.377	0.357	0.330	0.316	
	taud	2.353	2.384	2.293	2.238	2.291	2.294	2.302	2.355	2.411	2.442	2.405	2.335	
	Ebn at Noon	205	240	254	262	269	267	265	261	253	233	203	185	
	Edn at Noon	19	25	33	39	39	39	38	35	29	23	18	16	
	Edn at Noon	19	25	33	39	39	39	38	35	29	23	18	16	
(44) All-Sky Solar Radiation	RadAvg	215	434	804	1265	1574	1643	1584	1334	976	554	269	170	
	RadStd	21	61	81	130	116	126	148	91	82	43	27	18	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	+0.83	N/A	N/A	N/A	N/A	N/A	
(47) Regional (73 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page